INTRODUCTION

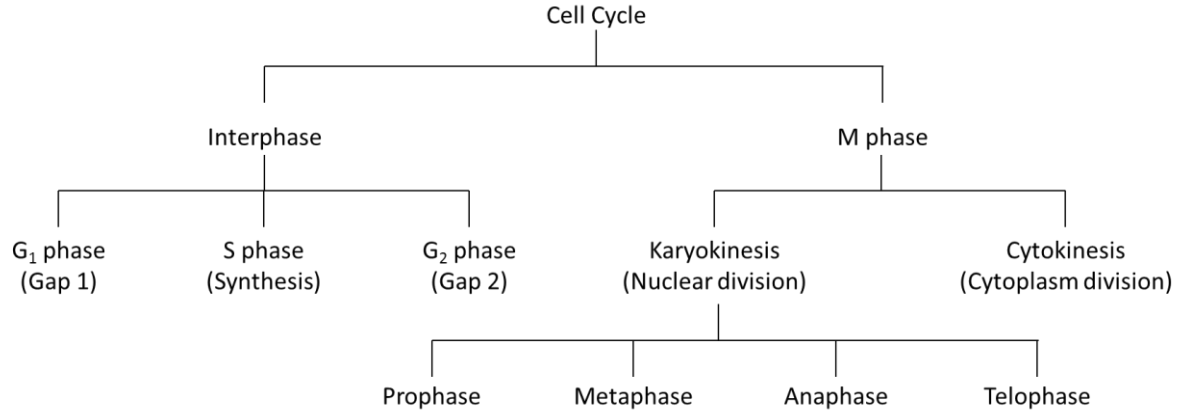
Are you aware that all organisms, even the largest, start their life from a single cell? You may wonder how a single cell then goes on to form such large organisms. **Growth and reproduction are characteristics of cells,** indeed of all living organisms. All cells reproduce by dividing into two daughter cells. These newly formed daughter cells can themselves grow and divide, giving rise to a new cell population that is formed by the growth and division of a single parental cells and its progeny. In other words, such cycles of growth and division allow a single cell to form a structure consisting of millions of cells.

CELL CYCLE

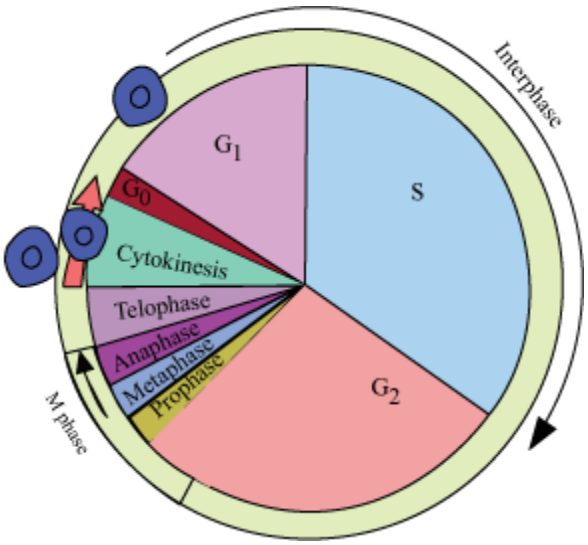
- The sequence of events by which a cell duplicates its genome, synthesises other cell constituents, and eventually divides into two daughter cells is termed as **cell cycle.**
- The events of the cell cycle are under genetic control.

Phases of Cell Cycle

- Duration of the cell cycle varies from organism to organism, and from cell to cell. Duration of the cell cycle in humans is 24 hrs, and in yeast is 90 min.
- Phases of the cell cycle are:



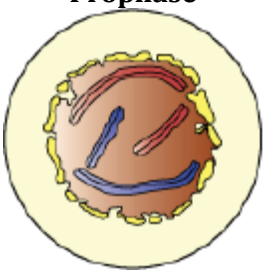
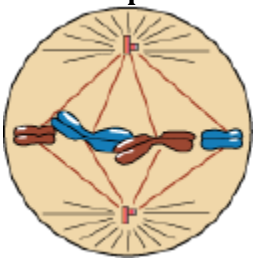
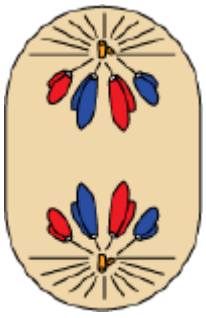
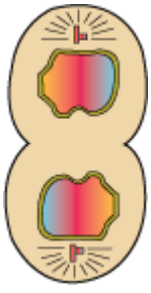
- **Interphase** represents the phase during which a cell prepares itself for division, grows and DNA is replicated in it. Interphase occupies more than 95% of the duration of the cell cycle.
- It is divided into 3 phases – G_1, S and G_2.
  - **G_1 (Gap1) phase** is the phase between the end of mitosis and the initiation of DNA replication. The growth of the cell takes place in this phase.
  - **S (synthesis) phase** is the phase of DNA replication. DNA content doubles (from 2C to 4C), but chromosome number remains the same. (remains diploid; 2*n* only). In this phase, the centriole also duplicates.
  - **G_2 (Gap2) phase** is the phase in which the proteins needed for mitosis are synthesised. Cell growth continues in this phase.

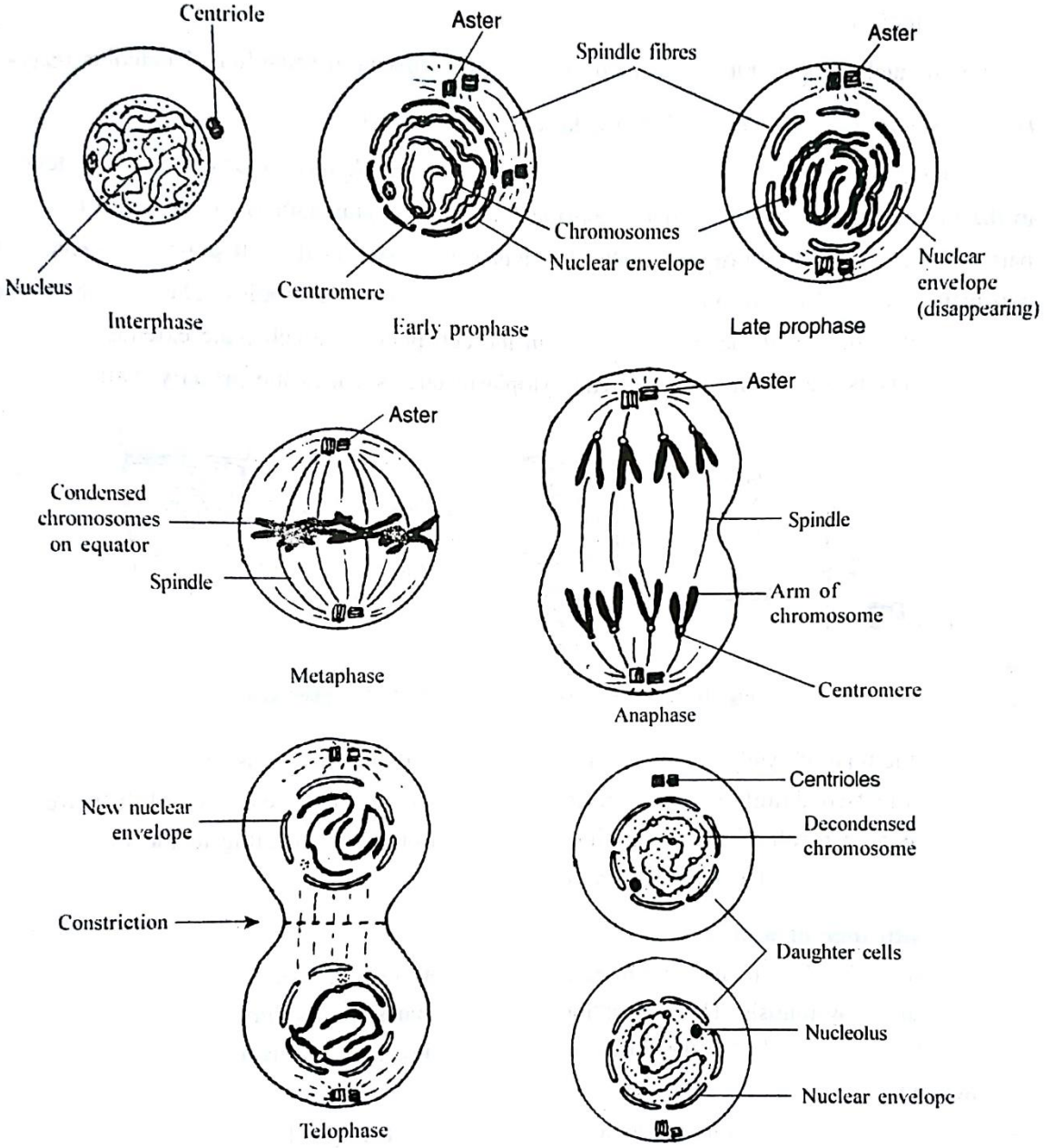


- **G_0 phase** is the quiescent stage in which the cells that do not divide enter after exiting from the **G_1 phase**. In this phase, the cells do remain metabolically active, but do not proliferate unless required.
- **M phase** represents the phase where the cell actually divides. It starts with nuclear division and ends with the division of the cytoplasm.

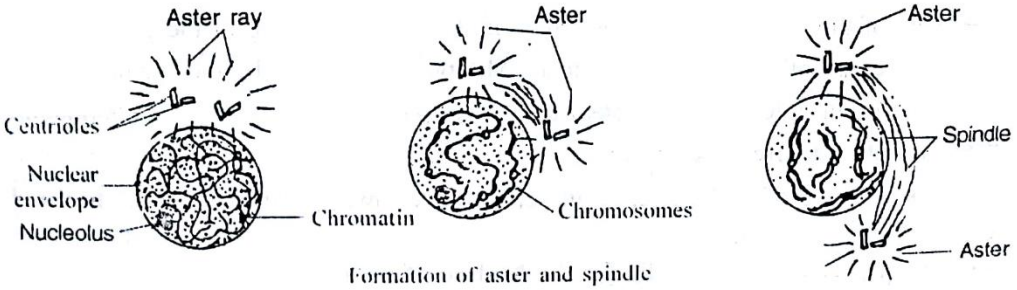
MITOSIS (M Phase)

- Also called **equational division** as number of chromosomes in parent and progeny remain the same.
- In animals mitosis occurs in somatic cells and in plants both haploid and diploid cells.
- Mitosis was first observed by **Stras Burger** (1870) in plant cells and later by **Flemming** (1879) in animal cell and also coined the term mitosis.
- Mitosis (M phase) is divided into 4 stages.
  1. **Prophase** (1st stage)
  2. **Metaphase**
  3. **Anaphase**
  4. **Telophase** (Last stage)

Phase	Characteristic features
Prophase 	• Follows S and G_2 phases of interphase • Nuclear envelope starts disintegrating. • Chromosome material condenses and untangles. • Centriole, which are duplicated during S phase, starts moving towards opposite poles. • Mitotic spindle starts assembling. • Microtubules appear.
Metaphase 	• Chromosomes condense completely. • Sister chromatids held together by centromere • At the surface of centriole, kinetochore appears. Here, spindle fibres attach. • Chromosomes lie at centre (equator) with each sister chromatid attached to spindle fibre of their respective poles with the help of kinetochore. [Alignment is called Metaphase plate]
Anaphase 	• Here, chromosome splits and daughter chromatids separate. • Chromatids move towards opposite poles. • Centromere leads and arms follow.
Telophase 	• Chromosomes reach at opposite poles. • Chromosomes start losing their individuality. • Chromatin material collects at poles. • Nuclear envelope appears around chromosome clusters. • Cell organelles such as ER, Golgi, reappear. • This phase is followed by interphase of next cycle.



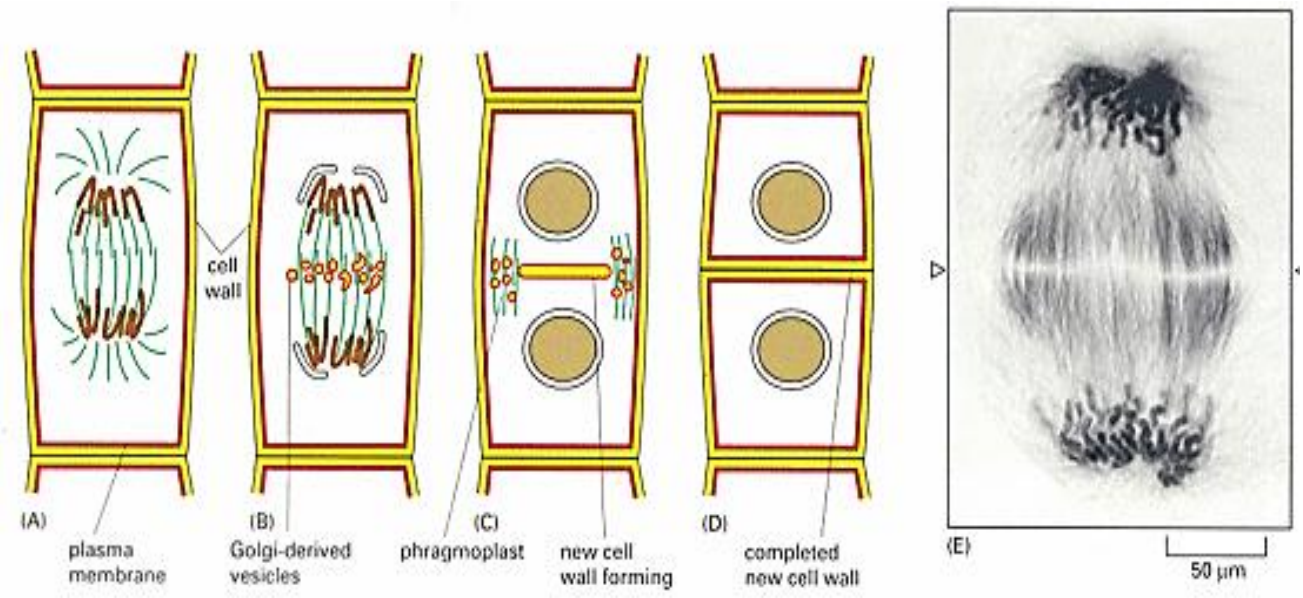
Different stages of mitosis in an animal cell



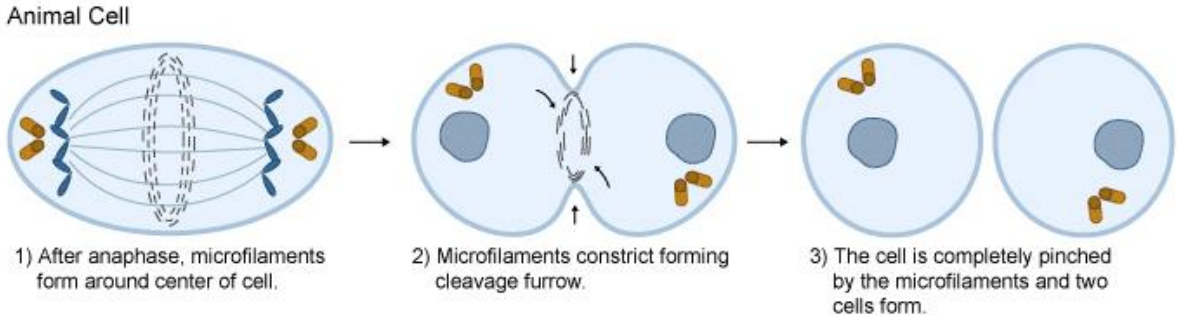
Formation of aster and spindle

Cytokinesis

- *Cyto* (cell) *kinesis* (division) follows the process of karyokinesis (nuclear division).
- Process of cytokinesis in **plant cell:**
    - New cell wall begins to form with precursor **cell plate** equivalent to middle lamella.
    - This new cell wall grows outwards.
    - Organelles such as mitochondria and plastids also get distributed between two daughter cells at the same time.
    - If cytokinesis does not follow karyokinesis, then cell becomes multinucleate, leading to the formation of **syncytium** (e.g., liquid endosperm in coconut).



- Process of cytokinesis in **animal cell:**
    - Furrow appears in the cell membrane.
    - This furrow deepens and joins in the centre to divide the cytoplasm.



SIGNIFICANCE OF MITOSIS

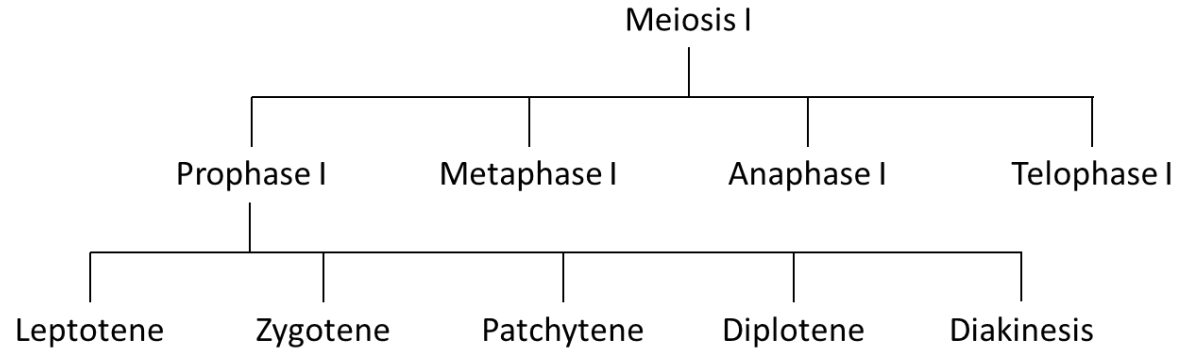
- Mitosis results in formation of diploid genetically identical daughter cells.
- Growth of the body (multicellular organism) takes place by mitosis.
- Cell repair and replacement of worn out tissues.
- Maintenance of nucleo-cytoplasmic ratio (it is the ratio of the size (volume) of the nucleus of a cell to the size of the cytoplasm of that cell
- Mitotic division in the meristematic tissue – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.
- Vegetative reproduction in plants takes place by mitosis.

MEIOSIS - events and significance

- Specialized kind of cell division that reduces the chromosome number by half (thus called reductional division) resulting in production of haploid daughter cells (gametes)
- Haploid gametes fuse in sexual reproduction to give rise to diploid cells.
- Meiosis I starts after DNA has been replicated in S phase. Meiosis I is followed by meiosis II at the end of which **four haploid cells** are formed.

MEIOSIS - I

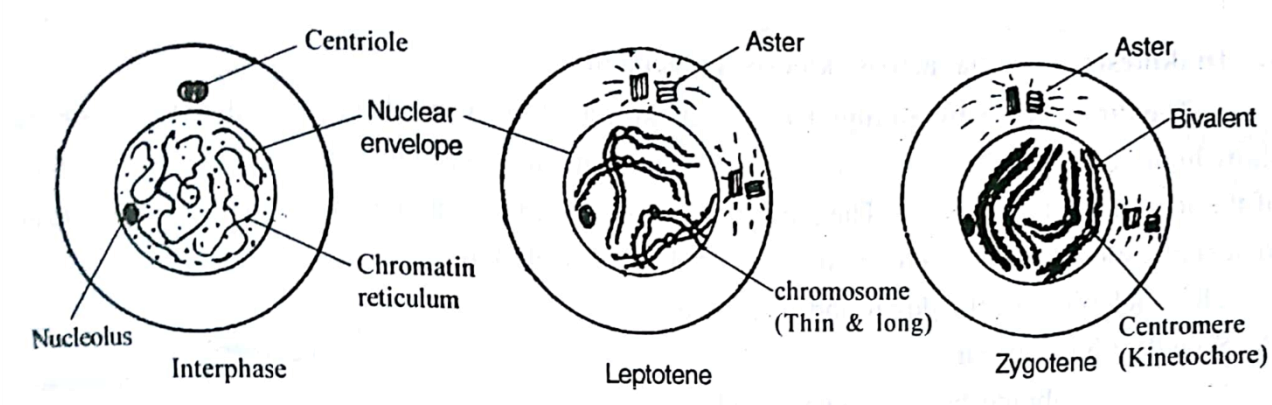
- Meiosis I is divided into 4 phases.

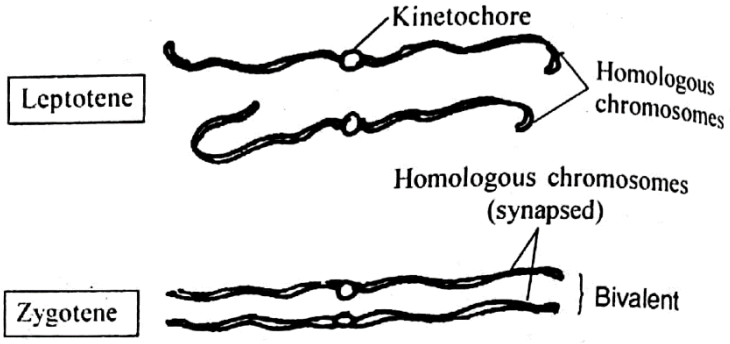


1. **Prophase I –** longest phase and is further sub-divided into 5 phases

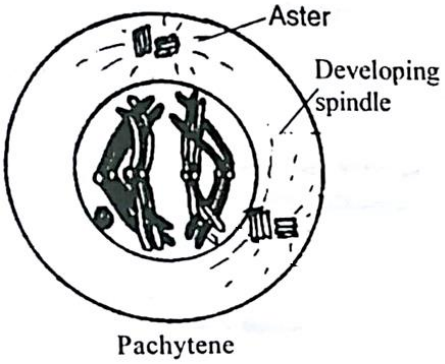
   i) ***Leptotene*** **–** Condensation makes chromosomes become distinct and compact.

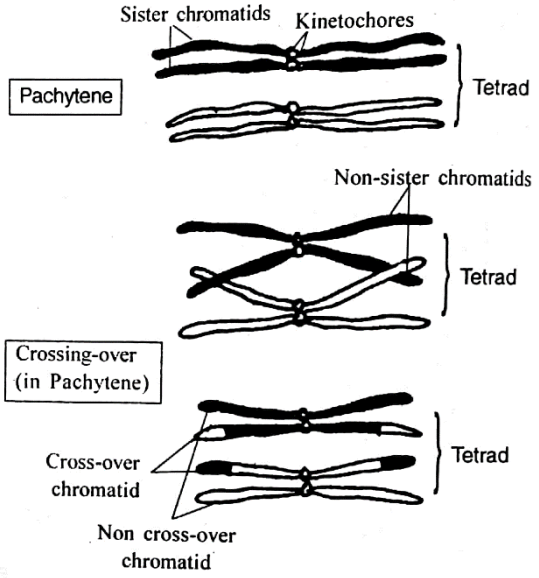
   ii) ***Zygotene*** **–** Homologous chromosomes start pairing together by a process called **synapsis** to form a complex structure called **synaptonemal complex**. Two synaptonemal complexes further form a complex called **bivalent of tetrad**.





iii) ***Pachytene* –** Longest phase of prophase I. Recombination nodules appear in this stage at the sites where crossing over has to take place between non-sister chromatids of homologous chromosomes.

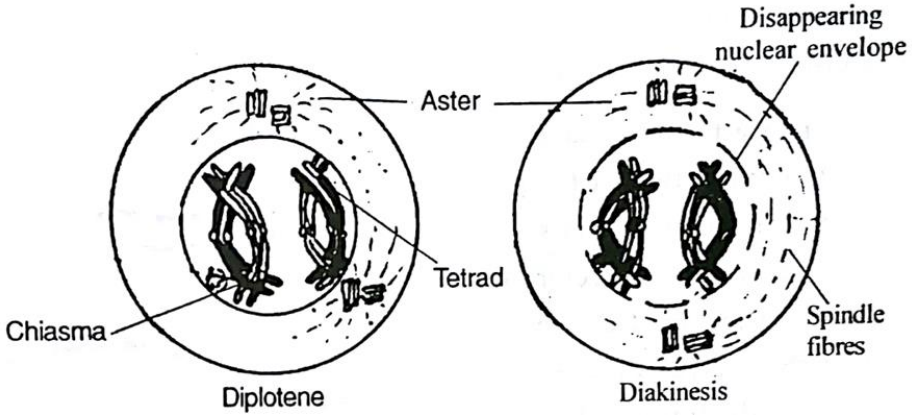




Actual reason for genetic difference in progenies
Crossing is mainly responsible for the genetic difference. Crossing over is the exchange of genetic material between two homologous chromosomes with the help of enzyme *recombinase*. It results in recombinant homologous chromosomes.

iv) ***Diplotene* –** Synaptonemal complex dissolves and recombinants separate from each other except at crossover sites to form X-shaped structure called **chiasmata**.

v) ***Diakinesis* –** Chiasmata terminalises and chromosomes condense. Mitotic spindle assembles and nucleolus and nuclear envelope disappear



2. **Metaphase I –** Bivalent chromosomes align on the equatorial plate and spindle fibres appear and attach to the homologous chromosomes.

3. **Anaphase I –** Homologous chromosomes separate; sister chromatids remain attached at their centromeres.

4. **Teleophase I –** Nuclear membranes and nucleolus reappear. Cytokinesis follows.

5. **Interkinesis –** Stage between meiosis I and meiosis II

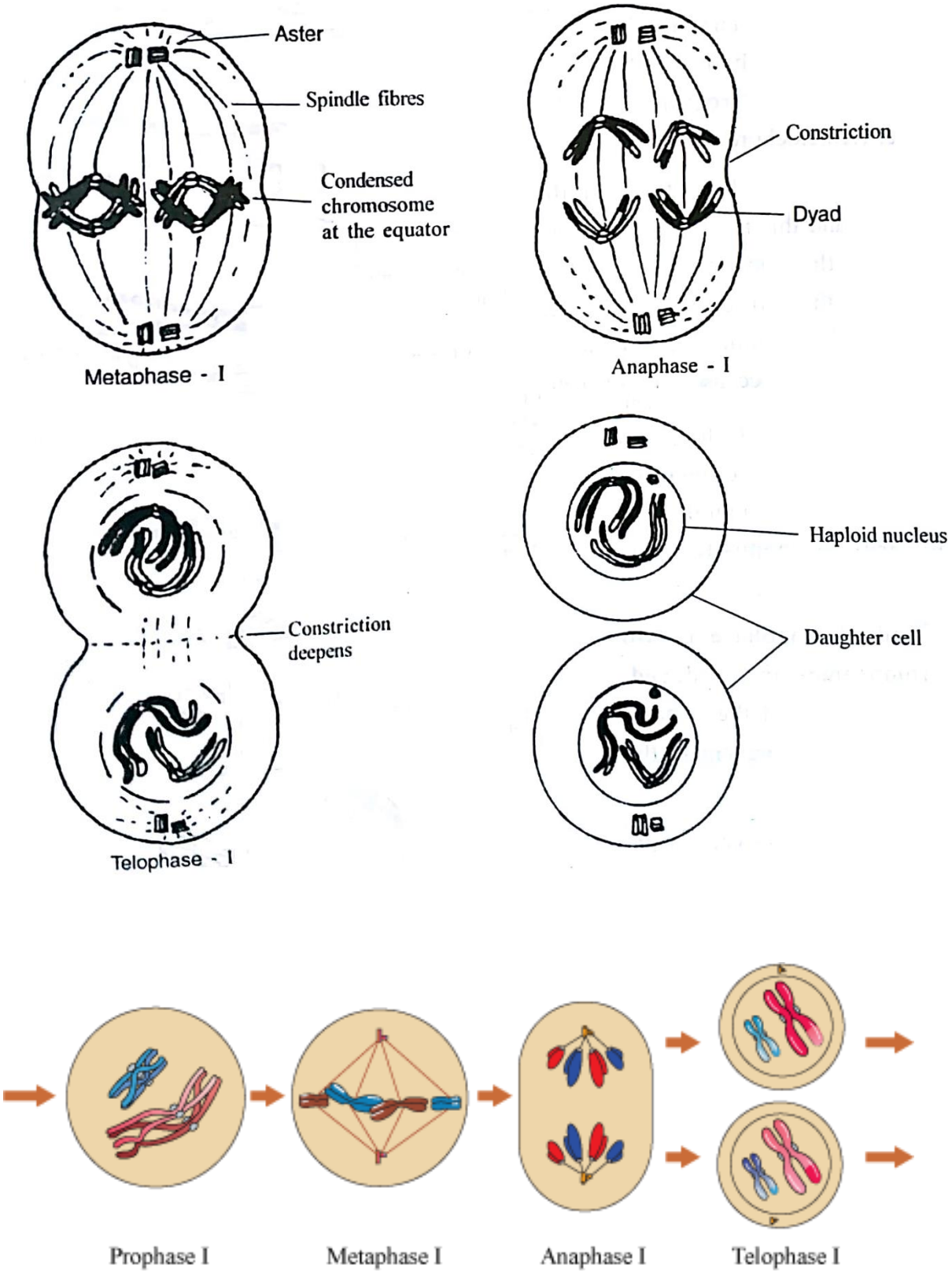


Figure: *NCERT diagram different stages of Meiosis I*

MEIOSIS - II – Resembles Mitosis

1. **Prophase II –** Chromosomes become compact and nuclear membrane disappears.
2. **Metaphase II –** Chromosomes align on equatorial plate and spindle fibres appear and attach to kinetochores of sister chromatids.
3. **Anaphase II –** Centromere of each chromosome splits and sister chromatids move towards opposite poles of cells.
4. **Telophase II –** Nuclear envelope reappears and cytokinesis follows, resulting in formation of a tetrad (4 haploid cells).

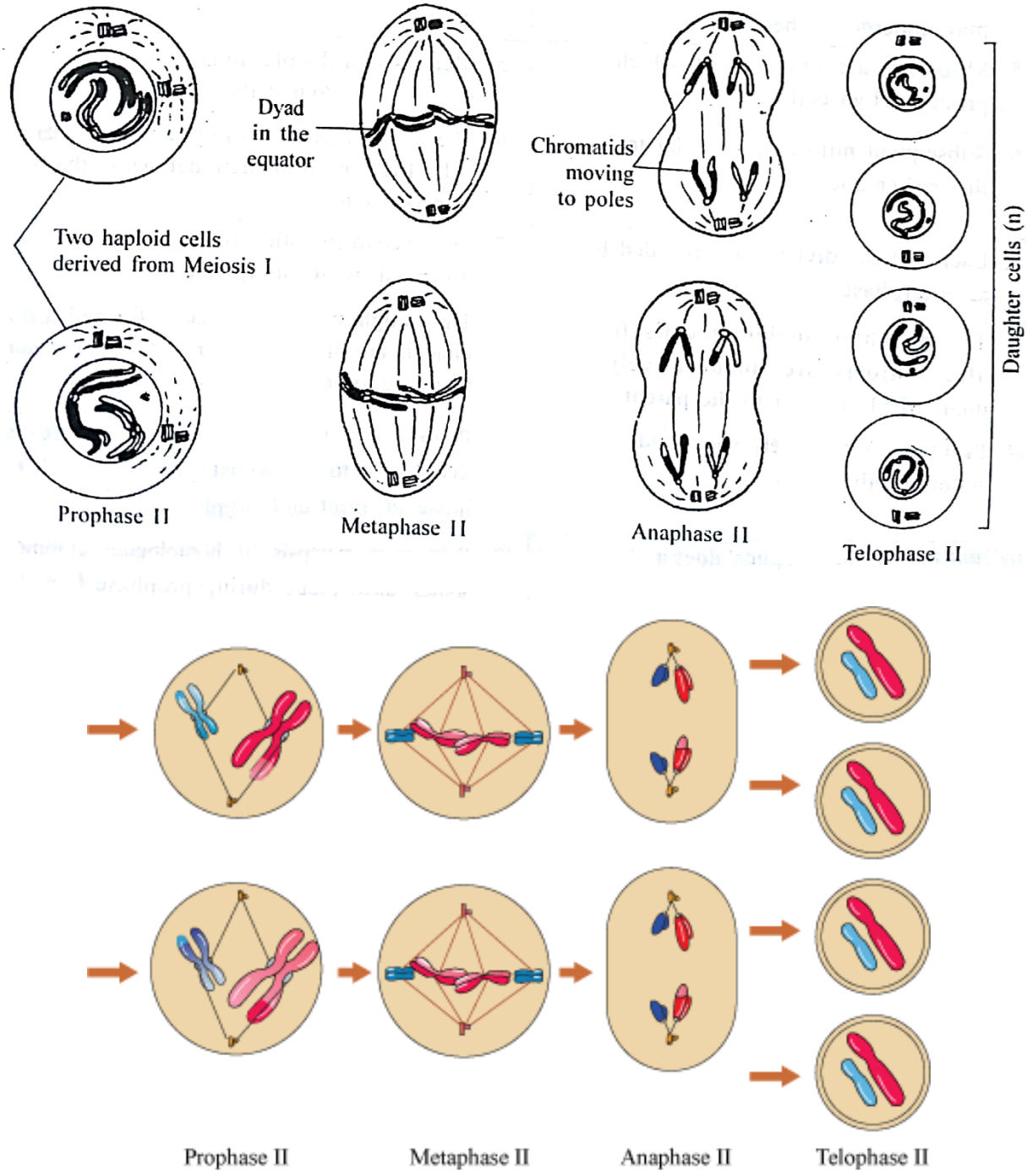


Figure: *NCERT diagram showing different stages of Meiosis II*

SIGNIFICANCE OF MEOSIS

- It results in reduction of chromosome number by half in gametes, which again doubles during fertilization. Therefore, it helps to conserve the chromosome number of species from generation to generation.
- Crossing-over, occurring in pachytene stage of meiosis I, is a source of genetic variability in sexually reproducing organisms.
- The variation thus formed serves as the raw material for evolution.